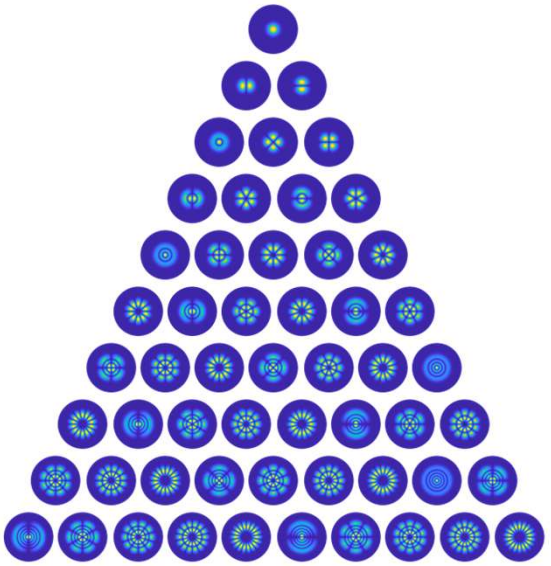
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

Lecture 11



Prof. Dr.-Ing. Georg Rademacher

Coherent Detection

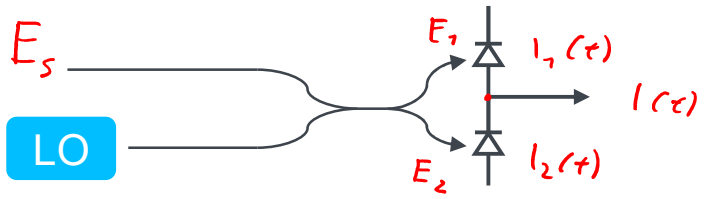
University of Stuttgart
Germany

$$E_s(t) = A_s(t) \exp(j\omega_s t)$$

$$E_{LO}(t) = A_{LO}(t) \exp(j\omega_{LO} t)$$

$$A_s(t) = \sqrt{P_s}\, e^{j\phi_s}$$

$$A_{LO} = \sqrt{P_{LO}}\, e^{j\phi_{LO}}$$



$$E_1 = \frac{1}{\sqrt{2}} (E_s + E_{LO})$$

$$E_2 = \frac{1}{\sqrt{2}} (E_s - E_{LO})$$

Coherent Detection

University of Stuttgart
Germany

$$I_1(t) = \frac{R}{2}\left[P_S + P_{LO} + 2\sqrt{P_S P_{LO}}\cos(\Delta\phi)\right]$$

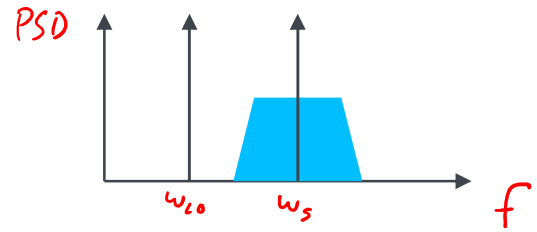
$$I_2(t) = \frac{R}{2}\left[P_S + P_{LO} - 2\sqrt{P_S P_{LO}}\cos(\Delta\phi)\right]$$

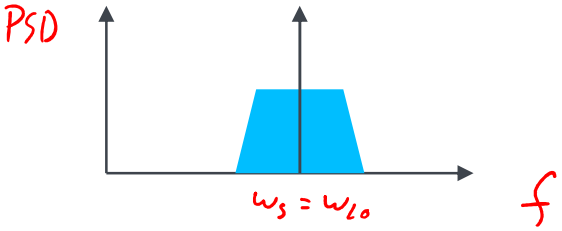
$$\Delta\phi = \phi_S - \phi_{LO}$$

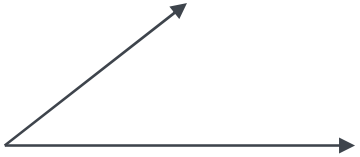
$$I(t) = I_1(t) - I_2(t) = 2R\sqrt{P_S P_{LO}}\cos(\Delta\phi)$$

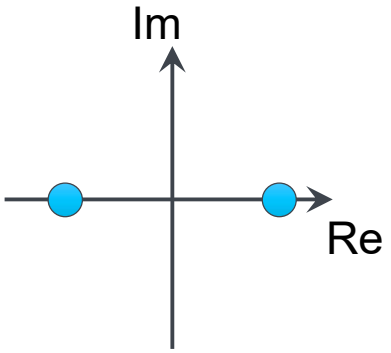
~~$I(t) =$~~

Homodyne Receivers

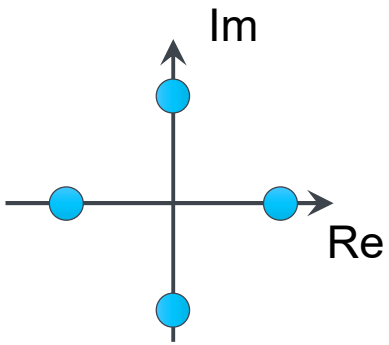


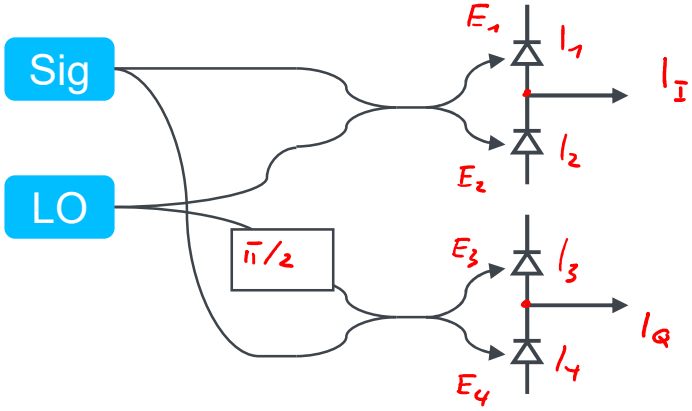






Homodyne Receivers: I and Q





$$E_1 = \frac{1}{2}(E_S + E_{LO})$$

$$E_2 = \frac{1}{2}(E_S - E_{LO})$$

$$E_3 = \frac{1}{2}(E_S + j\,E_{LO})$$

$$E_4 = \frac{1}{2}(E_S - j\,E_{LO})$$

$$I_1(t) = I_1 - I_2 = R\sqrt{P_S P_{LO}}\cos(\Delta\phi)$$

$$I_Q(t) = I_3 - I_4 = R\sqrt{P_S P_{LO}}\sin(\Delta\phi)$$

$$I_C(t) = I_1(t) + j I_Q(t) = R\sqrt{P_S(t) P_{LO}}\exp(j(\phi_S(t) + \phi_4(t)))$$

JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 34, NO. 1, JANUARY 1, 2016

Fundamentals of Coherent Optical Fiber Communications

Kazuro Kikuchi, *Fellow, IEEE, Fellow, OSA*

(Tutorial Review)

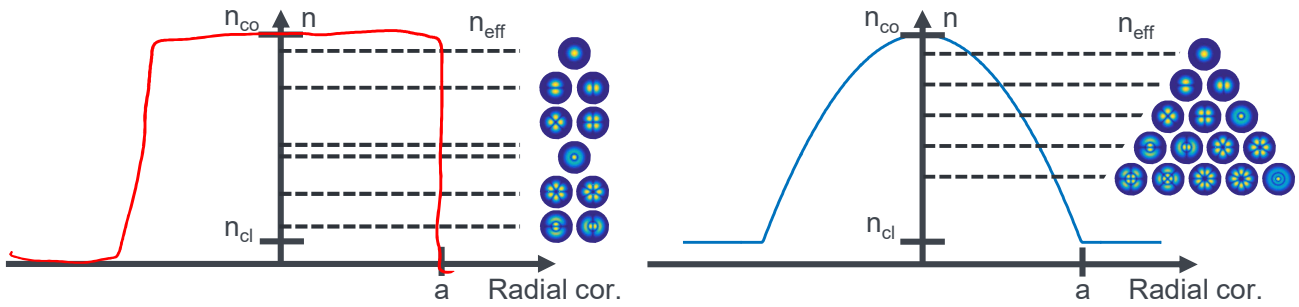
Bonus Test

Bonus test

University of Stuttgart
Germany

Why are graded-index multi-mode fibers favorable over step-index multi-mode fibers for data transmission?

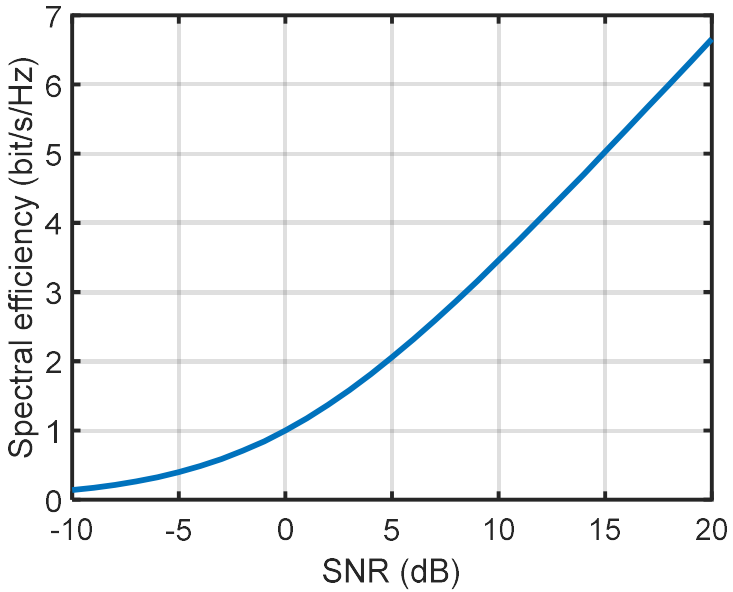
- **A** Lower loss
- **B** Lower differential mode delay
- **C** More modes



https://ilias3.uni-stuttgart.de/goto.php?target=xlvo_1_pin_QX5Z

Optical transmission link

University of Stuttgart
Germany



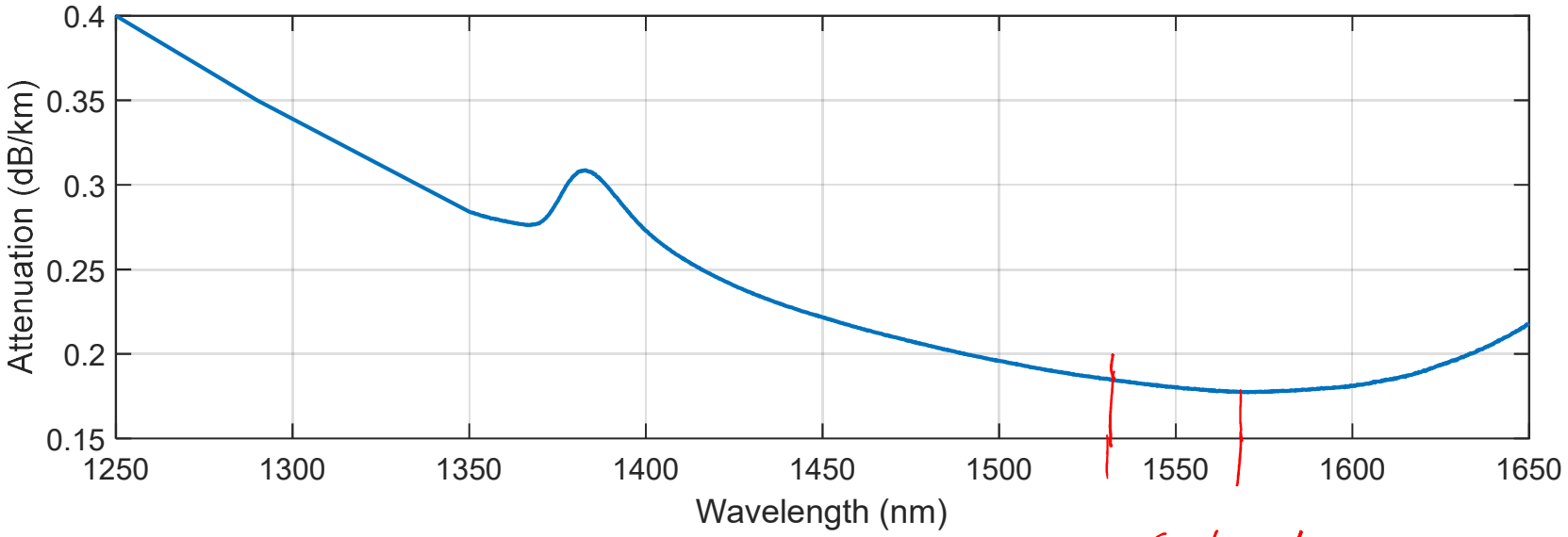
$x \rightarrow (+) \rightarrow y = x + n$, with n added at the $(+)$

$$C = B \cdot \overbrace{\log_2(1 + SNR)}^{\text{Spectral Efficiency [SE]}}$$

- B: Bandwidth
- SNR: $\frac{P_{sig}}{P_{NOISE}}$

Wavelength-division multiplex (WDM)

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Germany

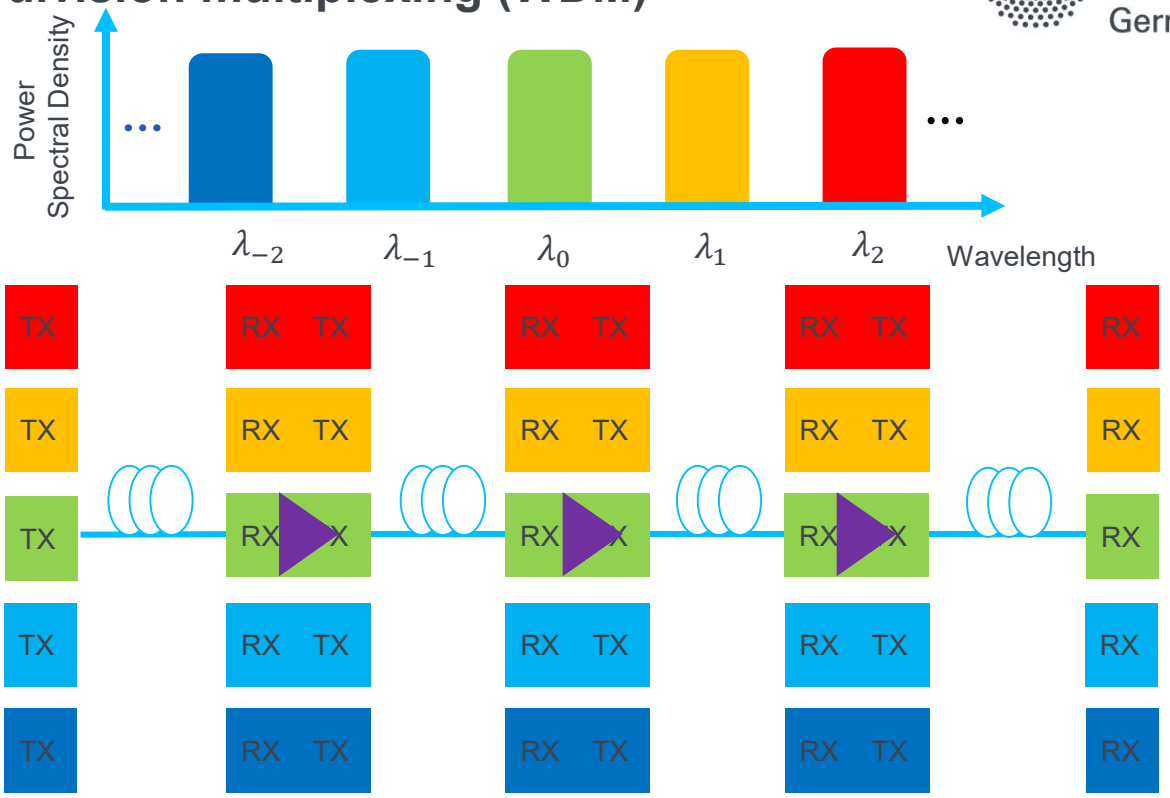


C-band (35nm, 4THz)

50-GHz spacing → 80 WDM channels

Wavelength division multiplexing (WDM)

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Germany





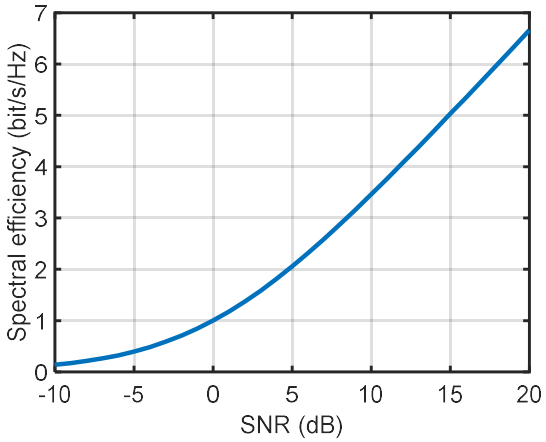
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Concept Test

Concept test

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Germany

An optical communications channel shall provide a data rate of 200 Gbit/s at a bandwidth of 50 GHz. What is the corresponding signal-to-noise ratio in dB?



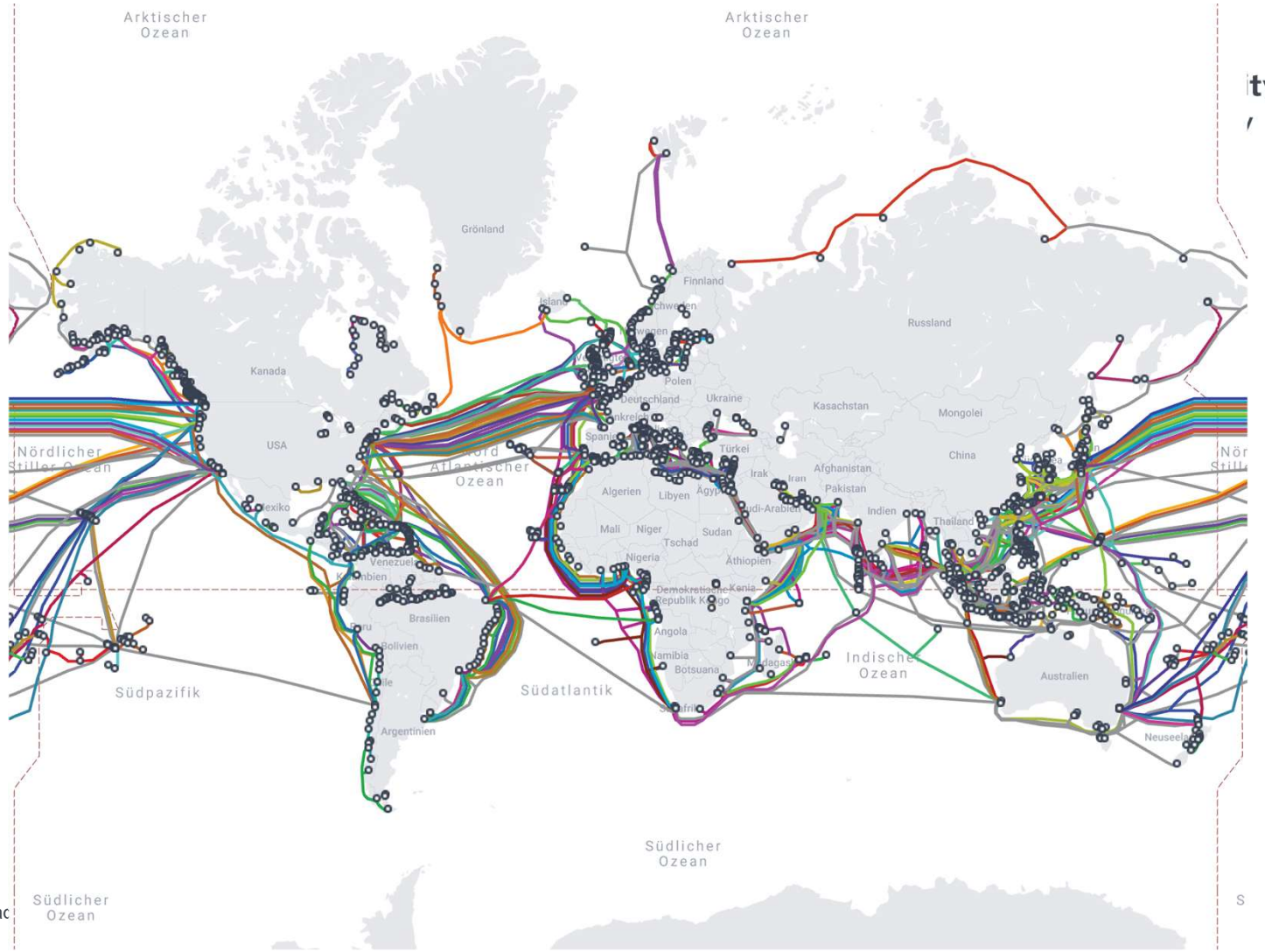
A 14.9 dB

B 8.5 dB

C 11.8 dB

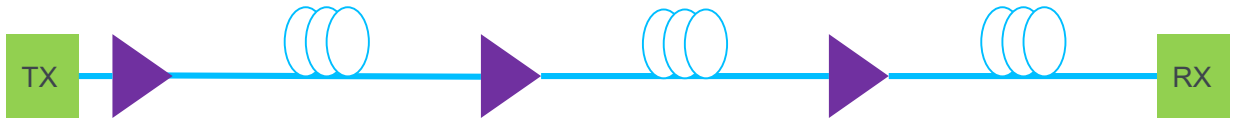
$$\text{Shannon Capacity} = B \cdot \log_2(1 + SNR)$$

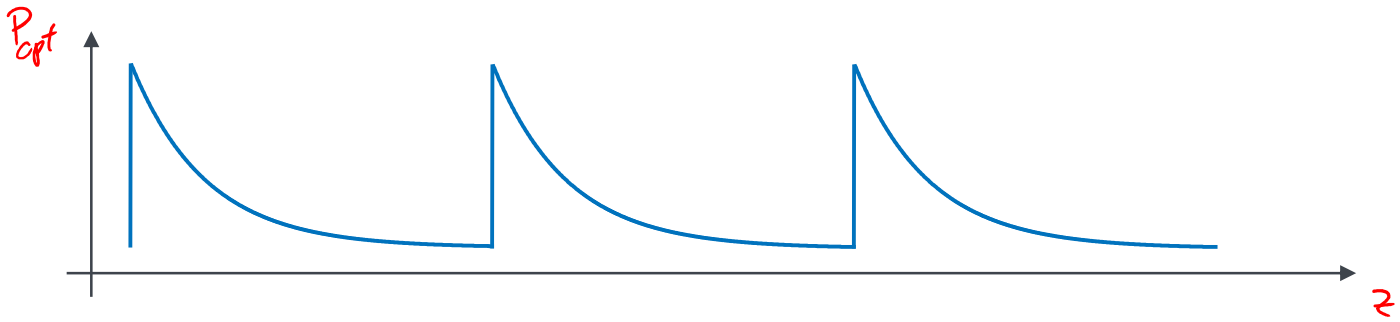
https://ilias3.uni-stuttgart.de/vote/E0X8


Arktischer Ozean
Arktischer Ozean
Grönland
Kanada
USA
Nördlicher Stiller Ozean
Nord Atlantischer Ozean
Mexiko
Venezuela
Kolumbien
Brasilien
Bolivien
Peru
Chile
Argentinien
Südpazifik
Südatlantik
Südlicher Ozean
Südlicher Ozean
Finnland
Schweden
Island
Norwegen
Russland
Polen
Deutschland
Frankreich
Spanien
Ukraine
Türkei
Kasachstan
Mongolei
China
Irak
Iran
Afghanistan
Pakistan
Algerien
Libyen
Ägypten
Saudi-Arabien
Indien
Thailand
Mali
Niger
Sudan
Tschad
Nigeria
Äthiopien
Kenia
Demokratische Republik Kongo
Angola
Namibia
Botsuana
Madagaskar
Indischer Ozean
Australien
Neuseeland

Power profile in amplified transmission link

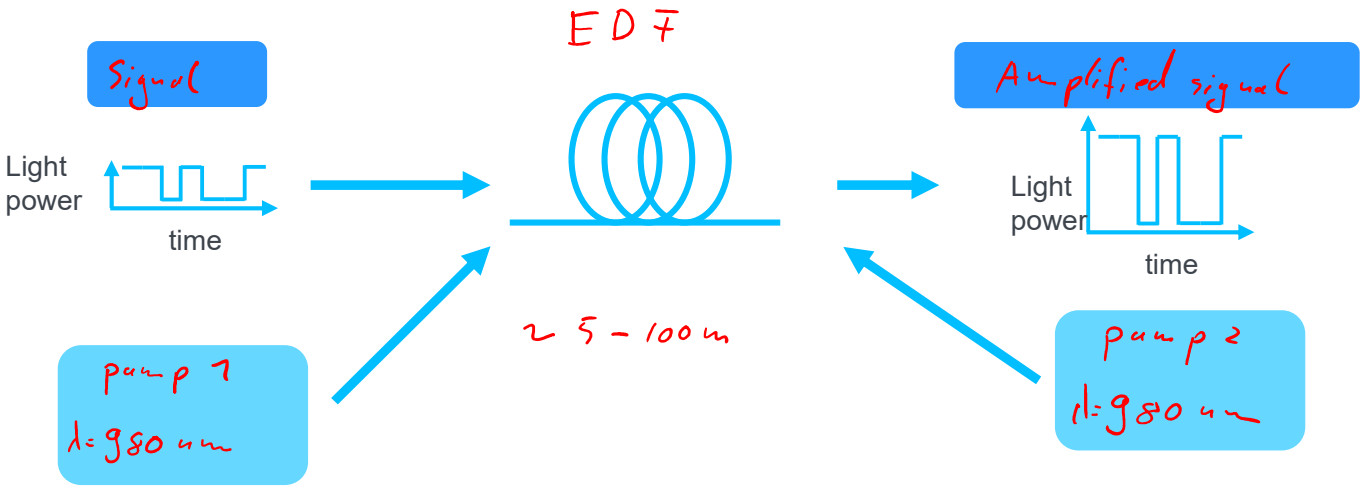
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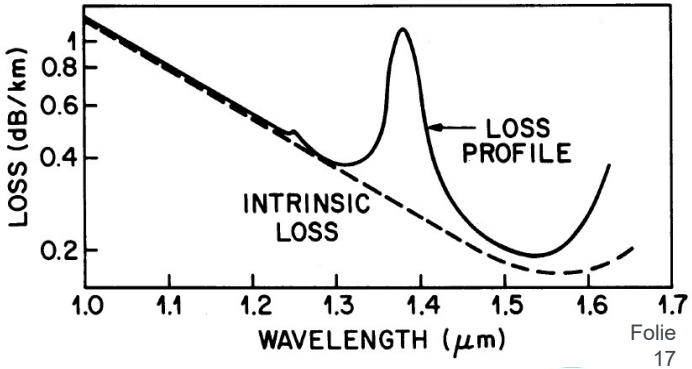




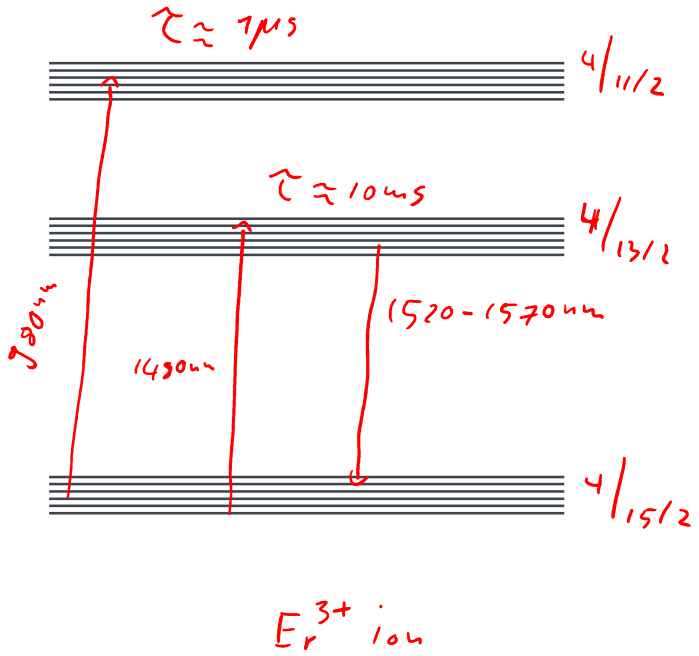
Erbium doped fiber amplifier (EDFA)

University of Stuttgart
Germany





Erbium-doped fiber amplifier (EDFA)



Pump and signal power evolution

University of Stuttgart
Germany

$$\frac{\partial P_s}{\partial z} = \Gamma_s \left(\sigma_s^e N_2 - \sigma_s^a N_1\right) P_s - \alpha P_s$$

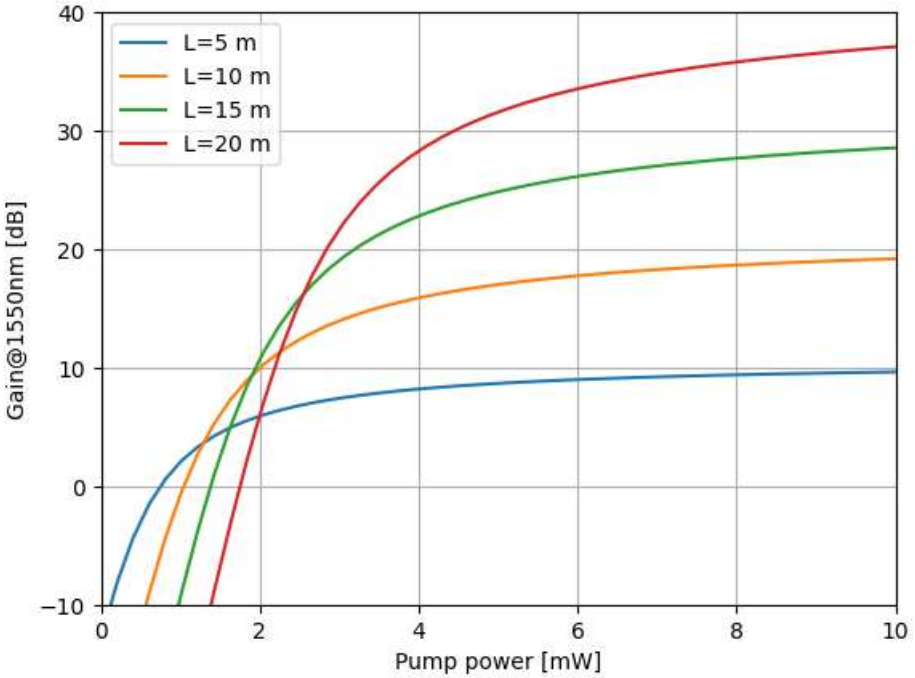
$$\frac{\partial P_p}{\partial z} = \Gamma_p \left(\sigma_p^e N_2 - \sigma_p^a N_1\right) P_p - \alpha' P_p$$

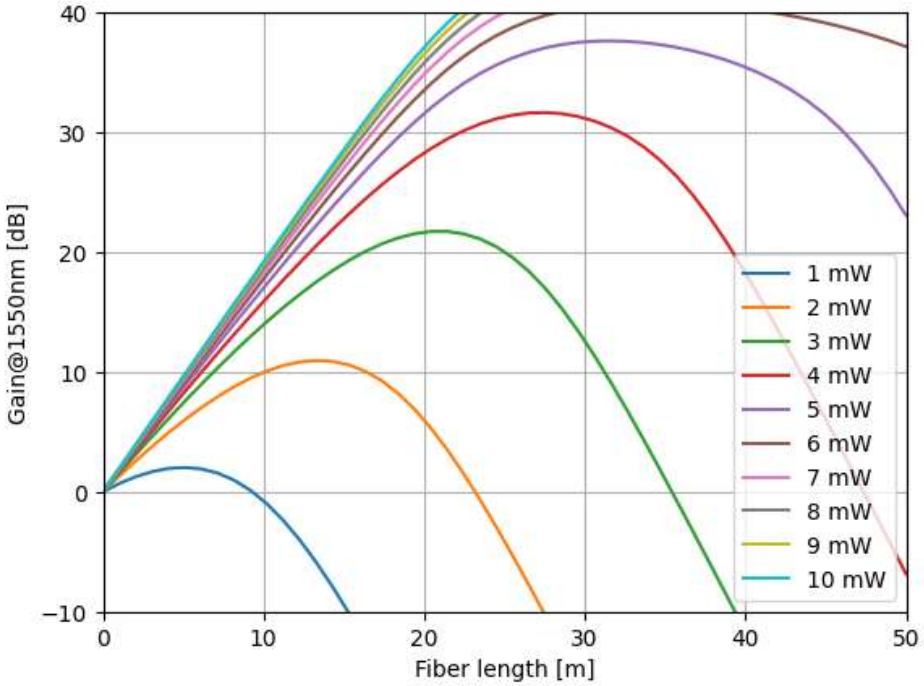
N_1, N_2 : populations in ground & excited state

α, α' : fiber attenuation

σ_j^a, σ_j^e absorption & emission cross section of frequency ω_j

Gain vs. pump power and fiber length





EDFA Noise

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Germany

$$NF = \frac{SNR_{in}}{SNR_{out}}$$

gain: G $\quad P_{out} = P_{in} \cdot G$

$$SNR_{in} = \frac{\langle I \rangle^2}{\sigma_s^2} = \frac{(R_D \cdot P_{in})^2}{2q(R_D \cdot P_{in})\Delta f} = \frac{P_{in}}{2h \cdot f \Delta f}$$

$P_{in} \cdot 2$ ⟵ ⟶ $P_{in} + 3dB$

$$R_D = \frac{q}{h \cdot f}$$

EDFA Noise

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Germany

$$SNR_{out} = \frac{(R_D G P_{in})^2}{\sigma^2} = \frac{G P_{in}}{(4 S_{ASE} + 2 h f) \Delta f}$$

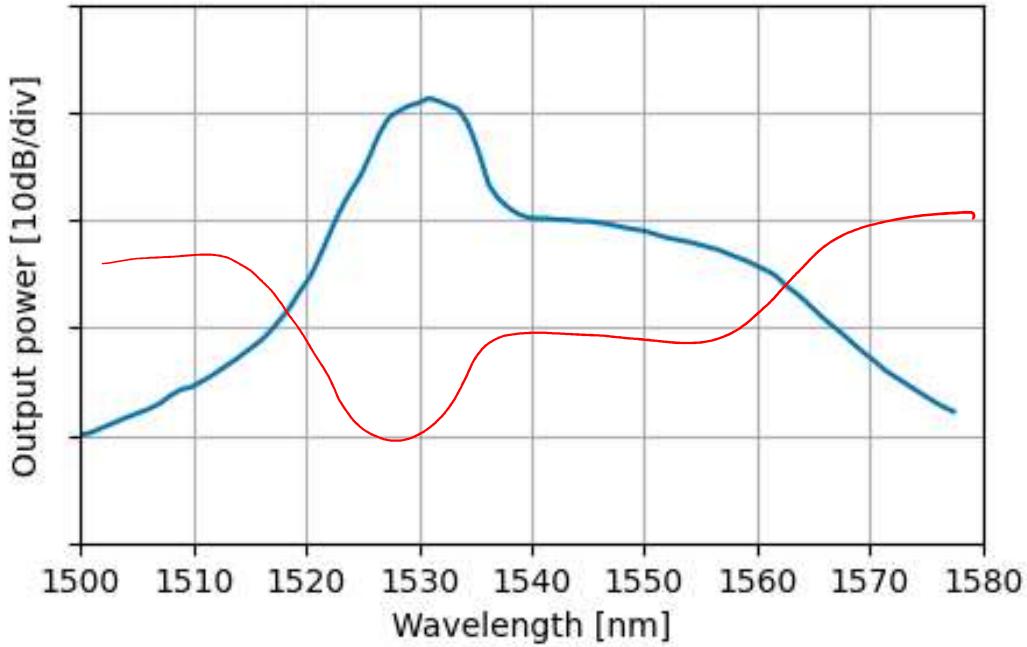
$$S_{ASE}(f) = n_{sp} h f_0 (G-1)$$

$$n_{sp} = \frac{\sigma_e N_2}{(\sigma_e N_2 - \sigma_a N_1)}$$

$$NF = 2 n_{sp} \left(1 - \frac{1}{G}\right) + \frac{1}{G} = \frac{1}{G}\left(1 + 2 \cdot n_{sp}(G-1)\right) \approx 2 \cdot n_{sp} \quad (G \gg 1)$$

$\rightarrow$ 3 dB for $n_{sp} = 1$

EDFA Noise


Output power [10dB/div]
1500
1510
1520
1530
1540
1550
1560
1570
1580
Wavelength [nm]



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Germany

Concept Test

Concept test

How is the SNR changed when doubling the input power into the optical receiver?



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Germany

A SNR reduces by 3 dB

B SNR increases by 2 dB

C SNR increases by 3 dB

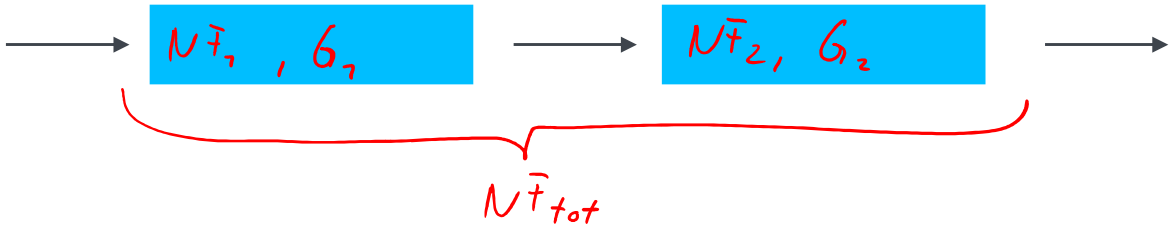
https://ilias3.uni-stuttgart.de/vote/NKE2

Noise in long distance transmission

2. Friis formula for two blocks

for Stuttgart

For two cascaded stages:



$$N\bar{F}_{Tot} = N\bar{F}_1 + \frac{N\bar{F}_2 - 1}{G_1}$$

Noise in long distance transmission

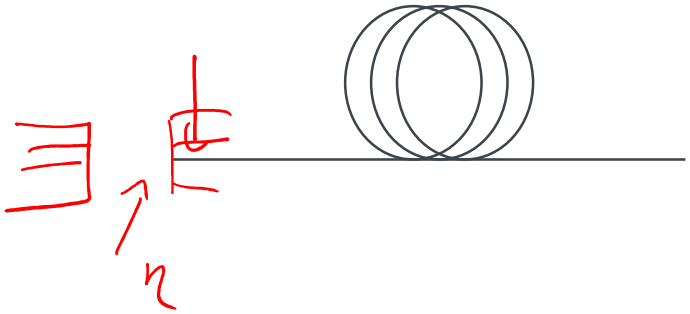
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Germany

Faserverstärker mit der Rauschzahl NF_1

Koppelwirkungsgrad η und damit einer Rauschzahl $1/\eta$ be

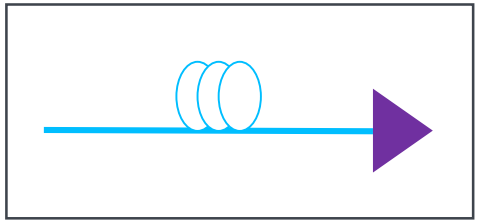
$$NF_2 = 2n_{sp}\left(1-\frac{1}{G}\right)+\frac{1}{G}$$

$$\overline{NF_1} = \frac{1}{G_1}$$

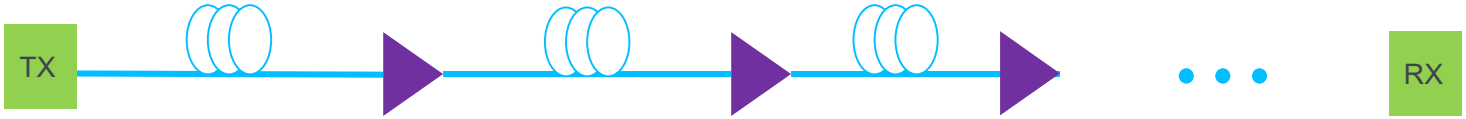


$$NF_{total} = \frac{1}{\eta} + \frac{NF_1 - 1}{\eta} = \frac{NF_1}{\eta}$$

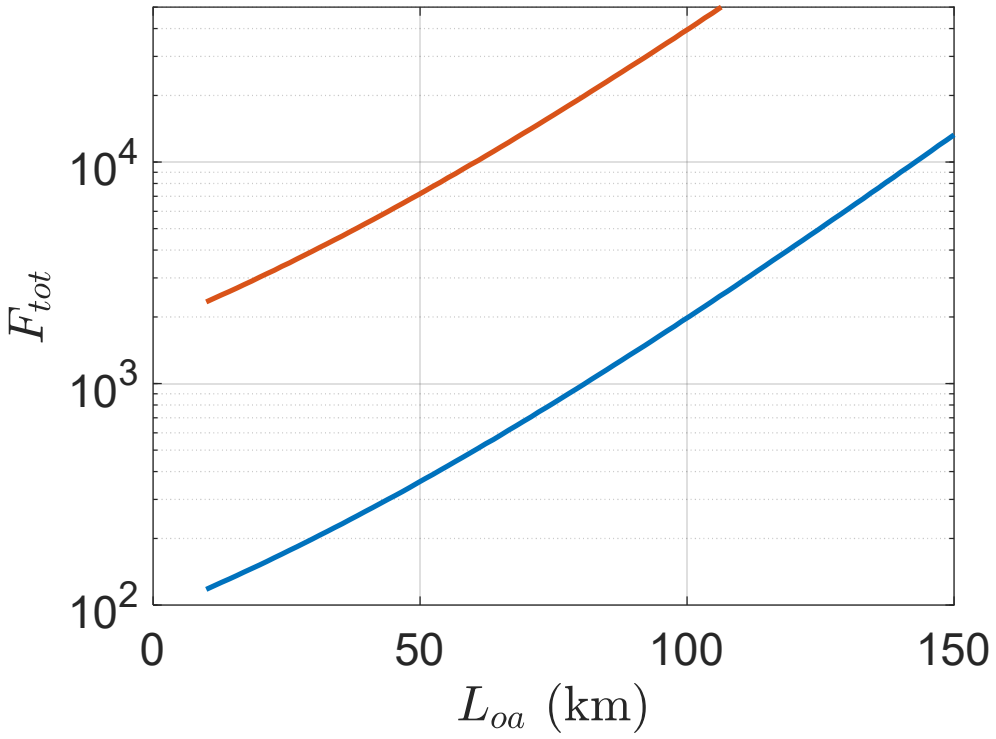
Noise in long distance transmission



Noise in long distance transmission



Noise in long distance transmission



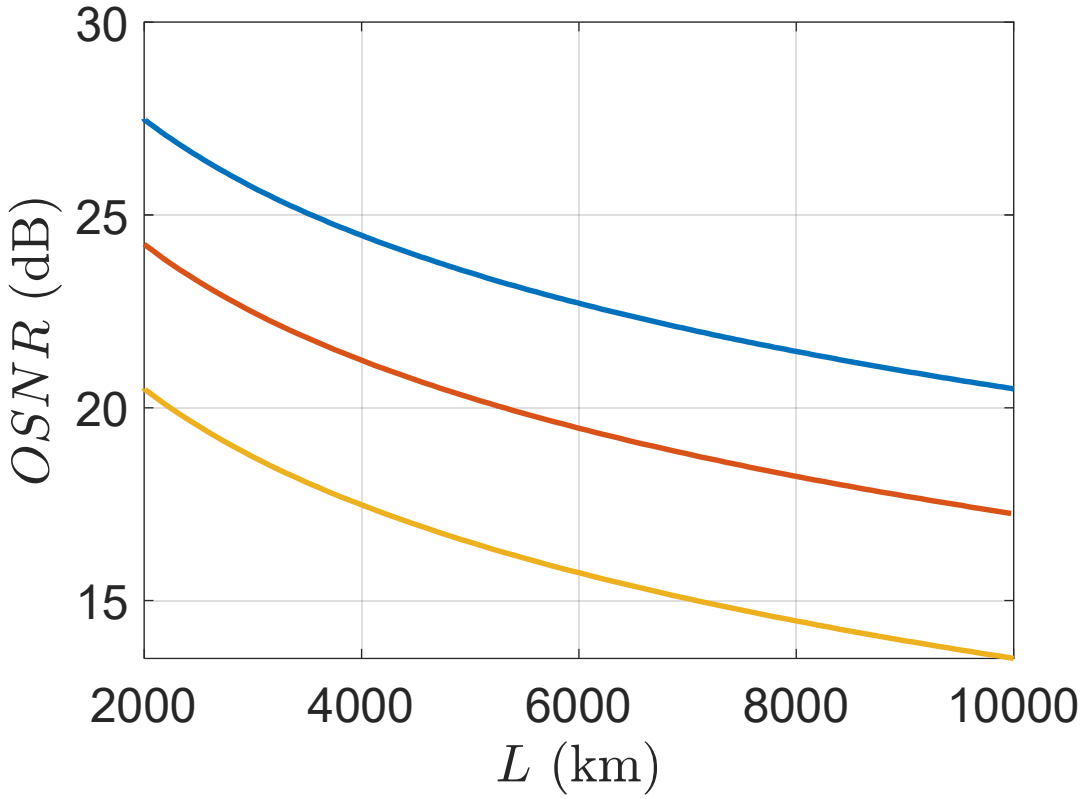
Optical Signal to Noise Ratio (OSNR)



University of Stuttgart
Germany

OSNR in long distance transmission

OSNR in long distance transmission



University of Stuttgart
Institute of electrical and optical communications

Thank you!



Prof. Dr.-Ing. Georg Rademacher

e-mail georg.rademacher@int.uni-stuttgart.de
phone +49 (0) 711 6856 7901
www. int.uni-stuttgart.de/

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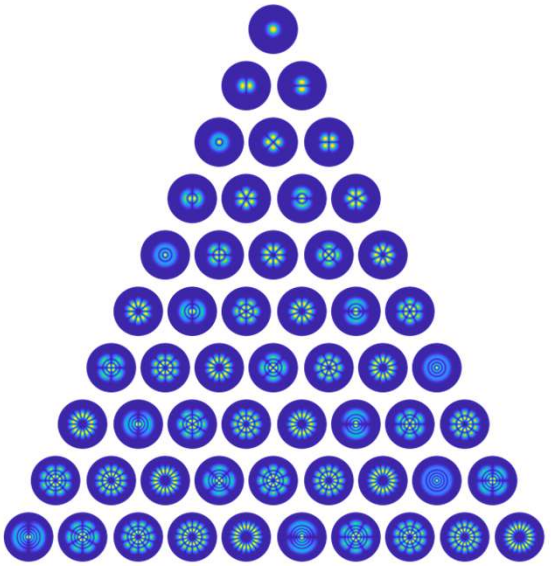


University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

Lecture 11



Prof. Dr.-Ing. Georg Rademacher

Coherent Detection

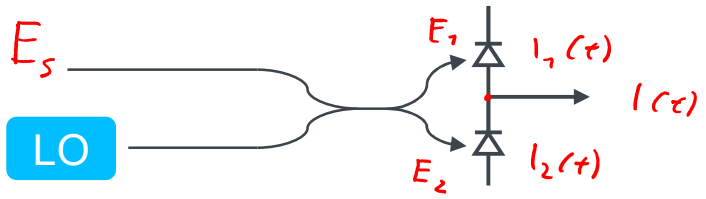
University of Stuttgart
Germany

$$E_s(t) = A_s(t) \exp(j\omega_s t)$$

$$E_{LO}(t) = A_{LO}(t) \exp(j\omega_{LO} t)$$

$$A_s(t) = \sqrt{P_s}\, e^{j\phi_s}$$

$$A_{LO} = \sqrt{P_{LO}}\, e^{j\phi_{LO}}$$



$$E_1 = \frac{1}{\sqrt{2}} (E_s + E_{LO})$$

$$E_2 = \frac{1}{\sqrt{2}} (E_s - E_{LO})$$

Coherent Detection

University of Stuttgart
Germany

$$I_1(t) = \frac{R}{2}\left[P_S + P_{LO} + 2\sqrt{P_S P_{LO}}\cos(\Delta\phi)\right]$$

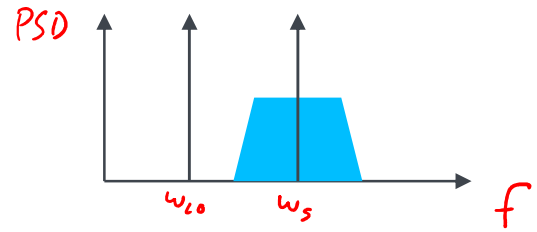
$$I_2(t) = \frac{R}{2}\left[P_S + P_{LO} - 2\sqrt{P_S P_{LO}}\cos(\Delta\phi)\right]$$

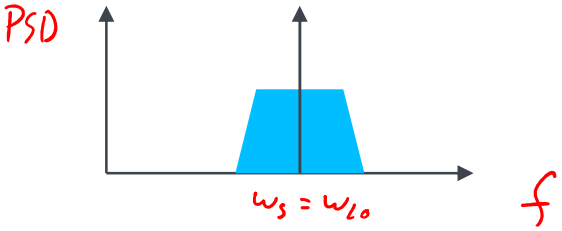
$$\Delta\phi = \phi_S - \phi_{LO}$$

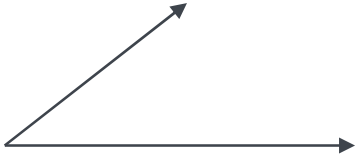
$$I(t) = I_1(t) - I_2(t) = 2R\sqrt{P_S P_{LO}}\cos(\Delta\phi)$$

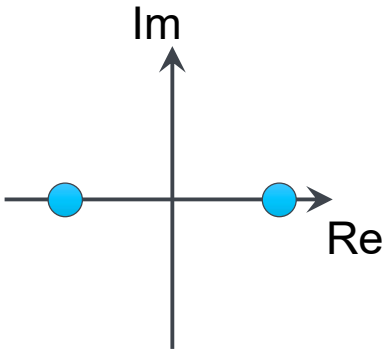
~~$I(t) =$~~

Homodyne Receivers

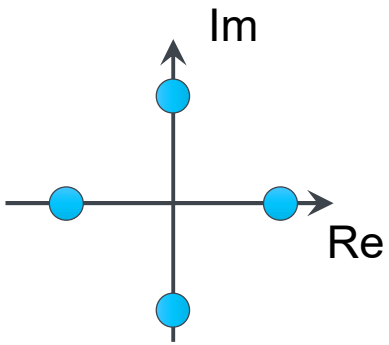


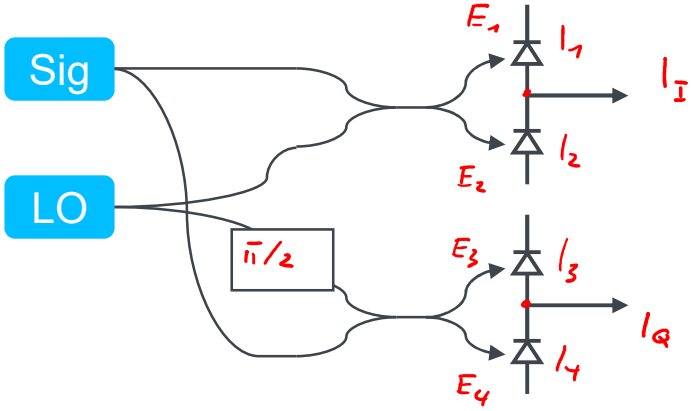






Homodyne Receivers: I and Q





$$E_1 = \frac{1}{2}(E_S + E_{LO})$$

$$E_2 = \frac{1}{2}(E_S - E_{LO})$$

$$E_3 = \frac{1}{2}(E_S + j\,E_{LO})$$

$$E_4 = \frac{1}{2}(E_S - j\,E_{LO})$$

$$I_1(t) = I_1 - I_2 = R\sqrt{P_S P_{LO}}\cos(\Delta\phi)$$

$$I_Q(t) = I_3 - I_4 = R\sqrt{P_S P_{LO}}\sin(\Delta\phi)$$

$$I_C(t) = I_1(t) + j I_Q(t) = R\sqrt{P_S(t) P_{LO}}\exp(j(\phi_S(t) + \phi_4(t)))$$

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Fundamentals of Coherent Optical Fiber Communications

Kazuro Kikuchi, *Fellow, IEEE, Fellow, OSA*

(Tutorial Review)

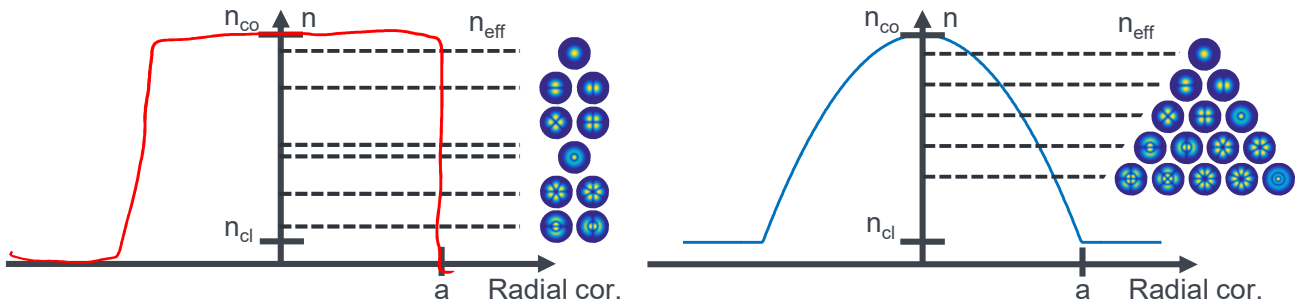
Bonus Test

Bonus test

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Why are graded-index multi-mode fibers favorable over step-index multi-mode fibers for data transmission?

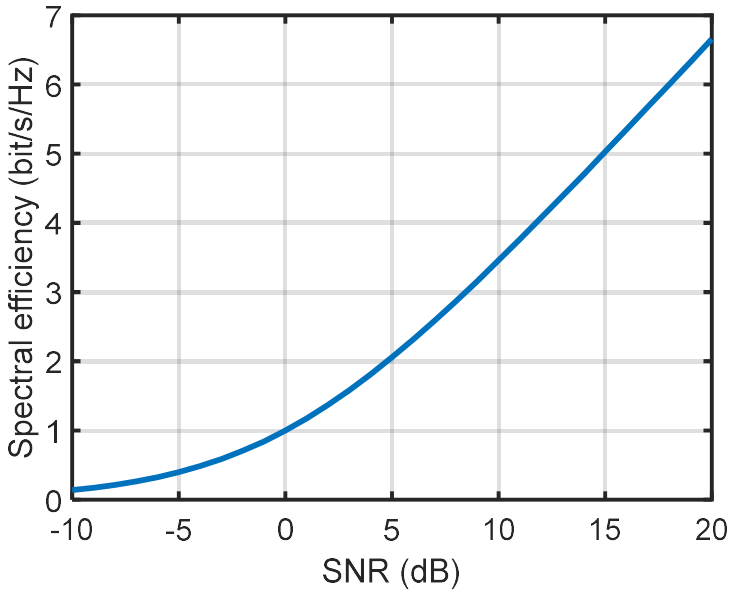
- **A** Lower loss
- **B** Lower differential mode delay
- **C** More modes



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Optical transmission link

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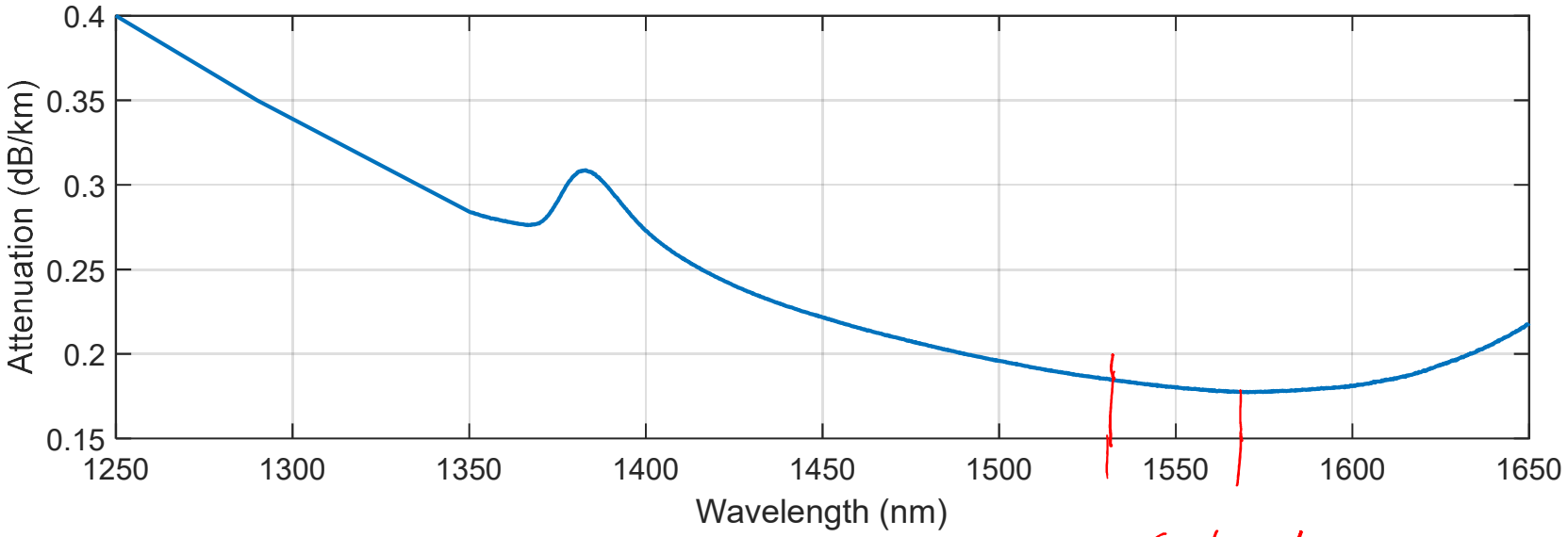


$x \rightarrow \oplus \rightarrow y = x + n$, with n added at $\oplus$

$$C = B \cdot \overbrace{\log_2(1 + SNR)}^{\text{Spectral Efficiency [SE]}}$$

- B: Bandwidth
- SNR: $\frac{P_{sig}}{P_{NOISE}}$

Wavelength-division multiplex (WDM)

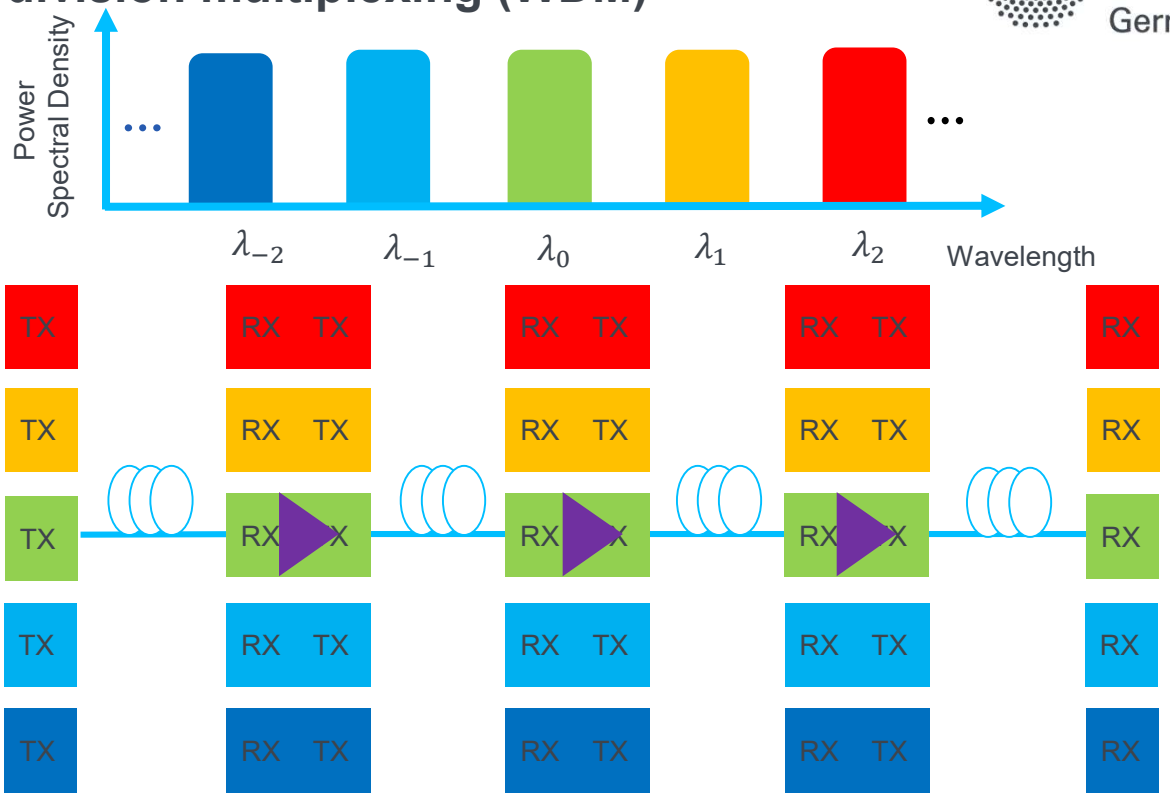


C-band (35nm, 4THz)

50-GHz spacing → 80 WDM channels

Wavelength division multiplexing (WDM)

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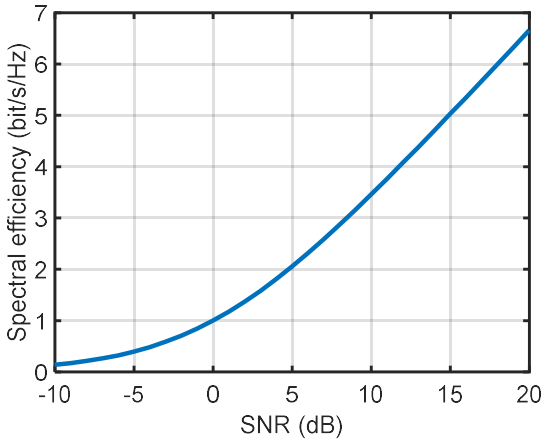
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Concept Test

Concept test

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An optical communications channel shall provide a data rate of 200 Gbit/s at a bandwidth of 50 GHz. What is the corresponding signal-to-noise ratio in dB?



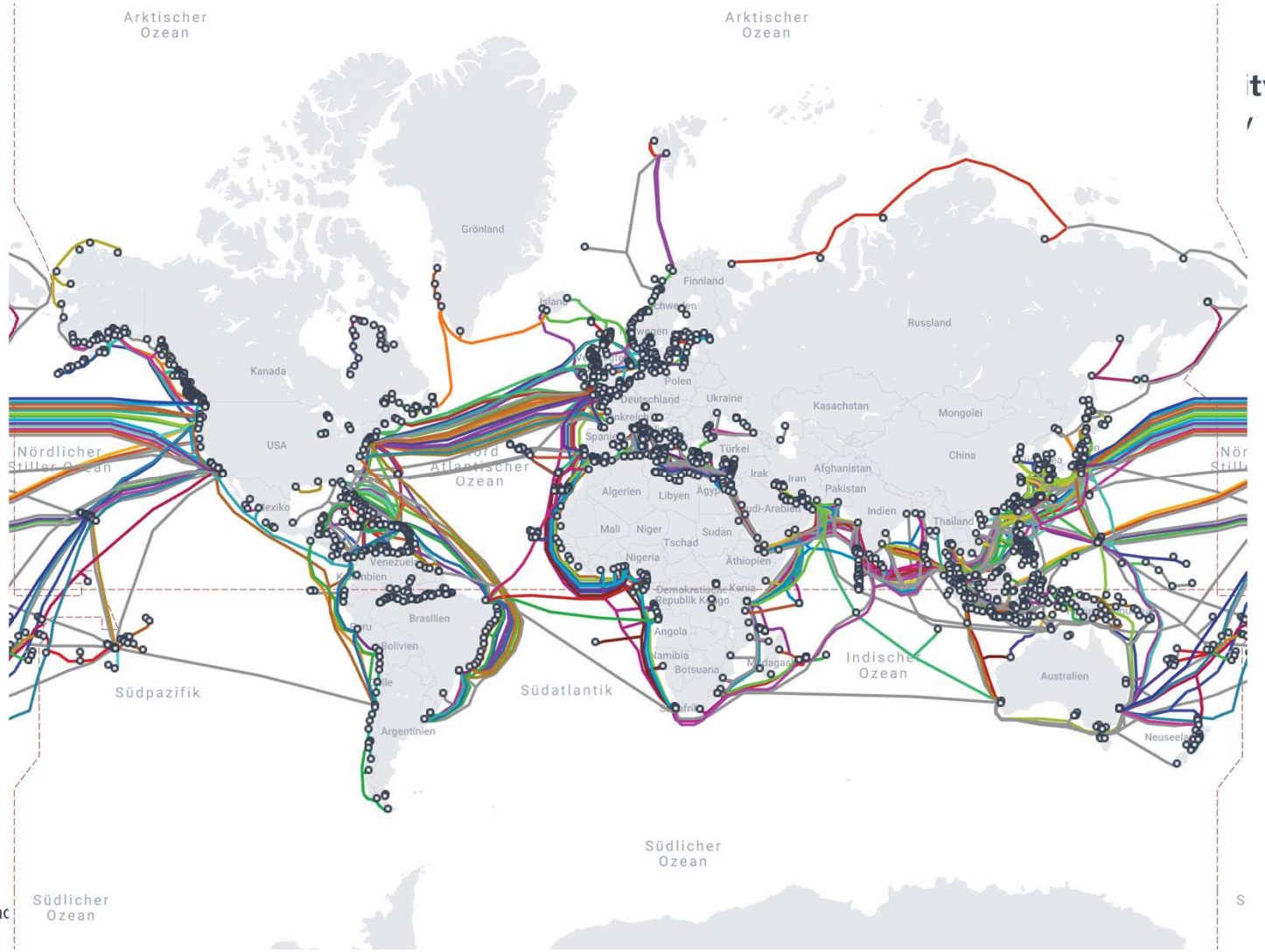
A 14.9 dB

B 8.5 dB

C 11.8 dB

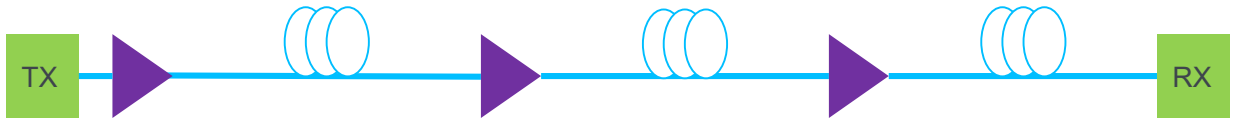
$$\text{Shannon Capacity} = B \cdot \log_2(1 + SNR)$$

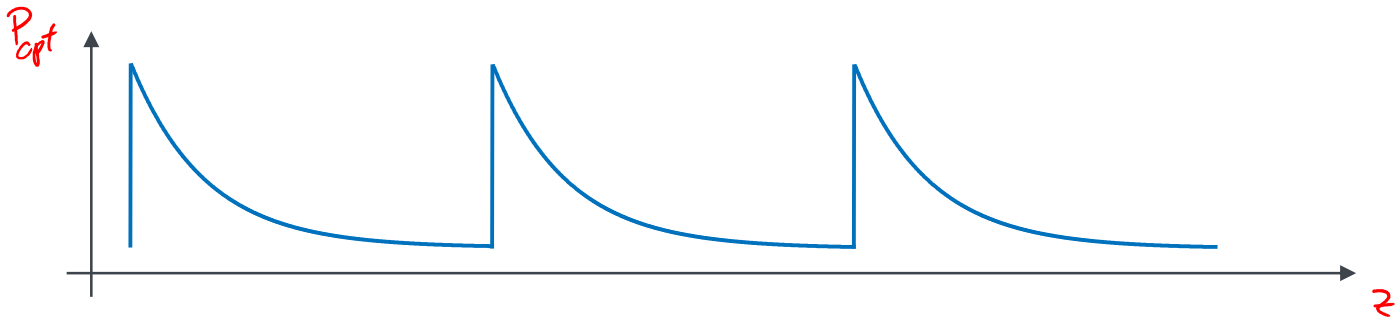
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Arktischer Ozean
Arktischer Ozean
Grönland
Kanada
USA
Nördlicher Stiller Ozean
Nord Atlantischer Ozean
Mexiko
Venezuela
Kolumbien
Brasilien
Bolivien
Peru
Chile
Argentinien
Südpazifik
Südatlantik
Südlicher Ozean
Südlicher Ozean
Finnland
Schweden
Island
Norwegen
Russland
Polen
Deutschland
Frankreich
Spanien
Ukraine
Türkei
Kasachstan
Mongolei
China
Irak
Iran
Afghanistan
Pakistan
Algerien
Libyen
Ägypten
Saudi-Arabien
Indien
Thailand
Mali
Niger
Sudan
Tschad
Nigeria
Äthiopien
Kenia
Demokratische Republik Kongo
Angola
Namibia
Botsuana
Madagaskar
Indischer Ozean
Australien
Neuseeland

Power profile in amplified transmission link

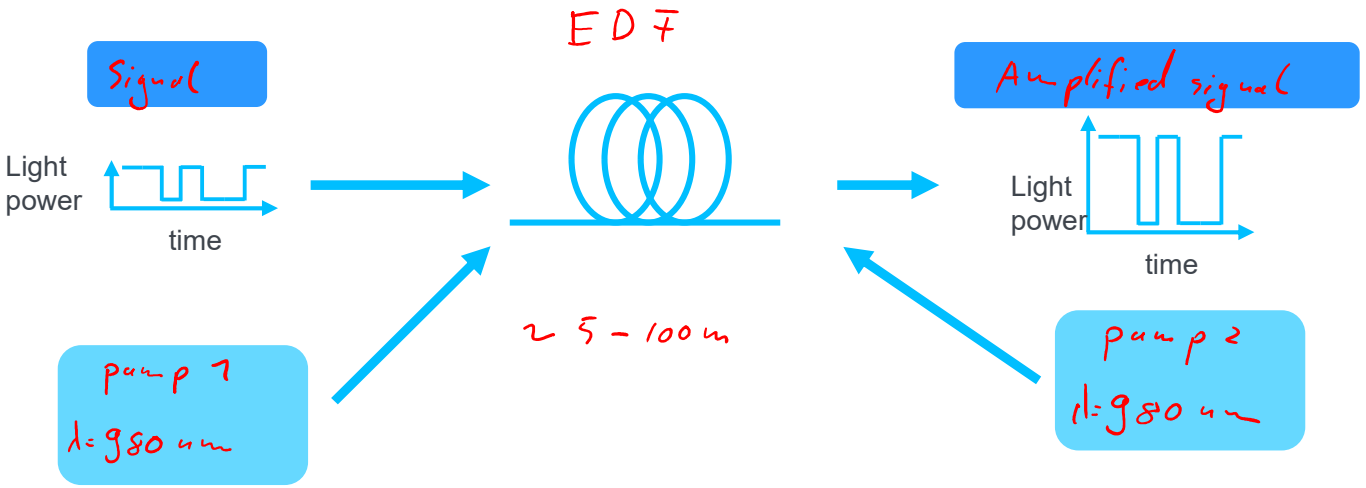
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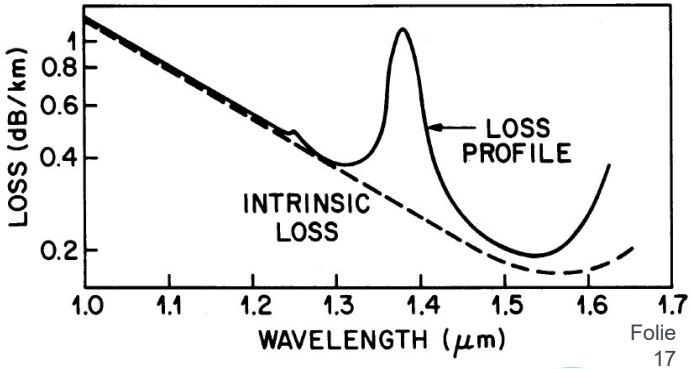




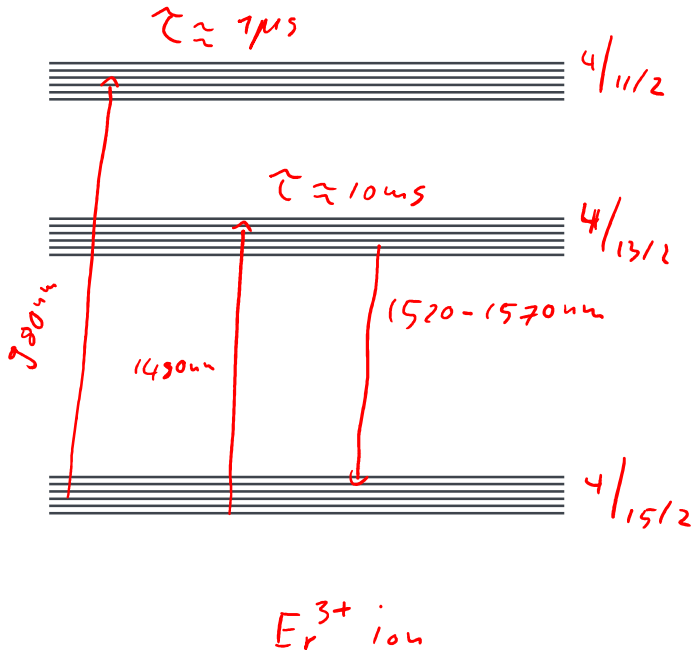
Erbium doped fiber amplifier (EDFA)

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Erbium-doped fiber amplifier (EDFA)



Pump and signal power evolution

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$$\frac{\partial P_s}{\partial z} = \Gamma_s \left(\sigma_s^e N_2 - \sigma_s^a N_1\right) P_s - \alpha P_s$$

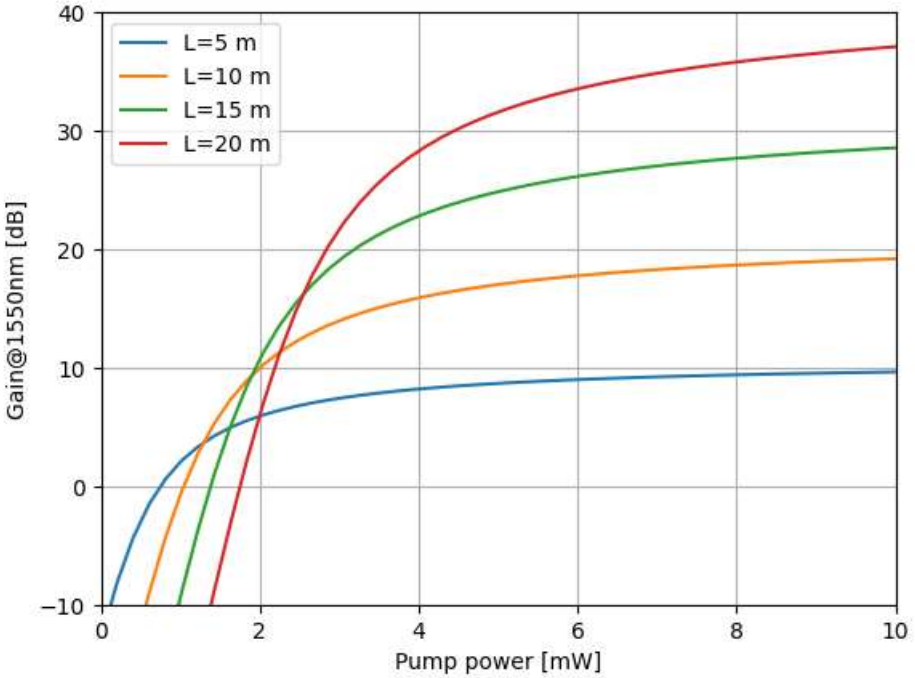
$$\frac{\partial P_p}{\partial z} = \Gamma_p \left(\sigma_p^e N_2 - \sigma_p^a N_1\right) P_p - \alpha' P_p$$

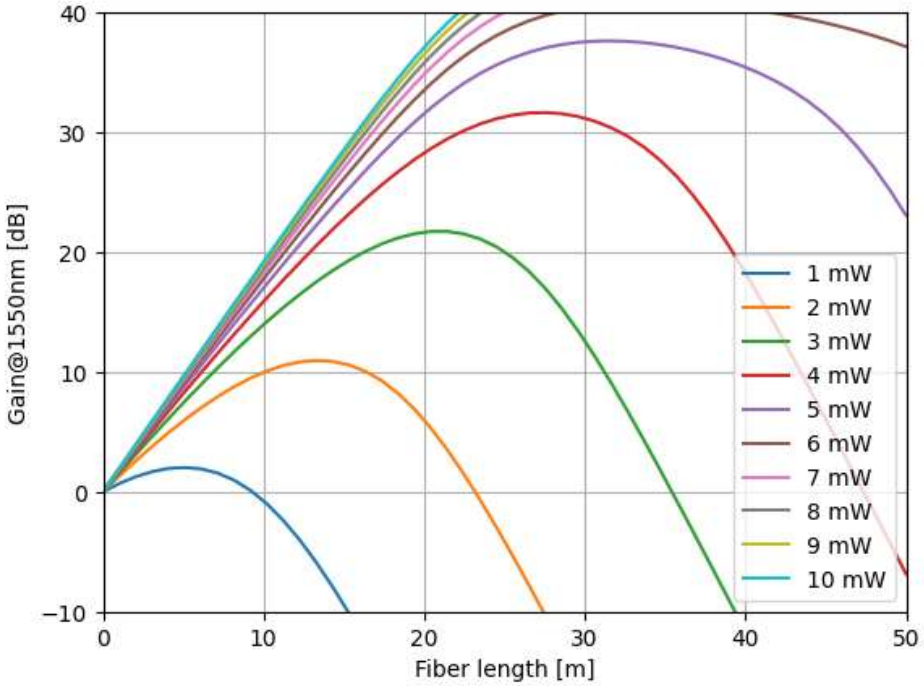
N_1, N_2 : populations in ground & excited state

α, α' : fiber attenuation

σ_j^a, σ_j^e absorption & emission cross section of frequency ω_j

Gain vs. pump power and fiber length





EDFA Noise

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$$NF = \frac{SNR_{in}}{SNR_{out}}$$

gain: G $\quad P_{out} = P_{in} \cdot G$

$$SNR_{in} = \frac{\langle I \rangle^2}{\sigma_s^2} = \frac{(R_D \cdot P_{in})^2}{2q(R_D \cdot P_{in})\Delta f} = \frac{P_{in}}{2h \cdot f \Delta f}$$

$P_{in} \cdot 2$ ⟵ ⟶ $P_{in} + 3dB$

$$R_D = \frac{q}{h \cdot f}$$

EDFA Noise

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$$SNR_{out} = \frac{(R_D G P_{in})^2}{\sigma^2} = \frac{G P_{in}}{(4 S_{ASE} + 2 h f) \Delta f}$$

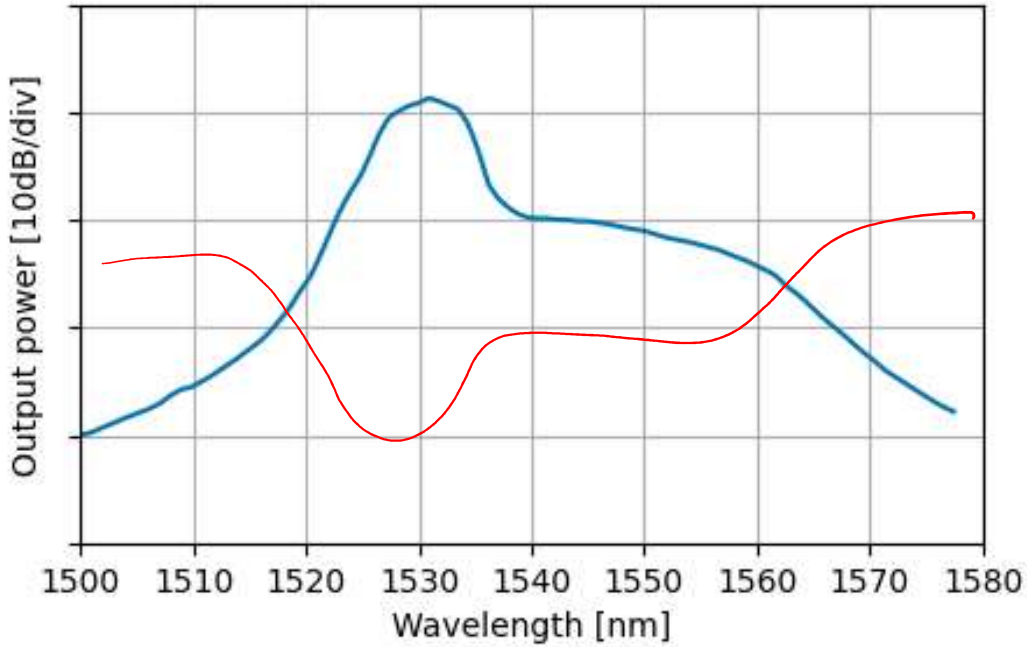
$$S_{ASE}(f) = n_{sp} h f_0 (G-1)$$

$$n_{sp} = \frac{\sigma_e N_2}{(\sigma_e N_2 - \sigma_a N_1)}$$

$$NF = 2 n_{sp} \left(1 - \frac{1}{G}\right) + \frac{1}{G} = \frac{1}{G}\left(1 + 2 \cdot n_{sp}(G-1)\right) \approx 2 \cdot n_{sp} \quad (G \gg 1)$$

$\rightarrow$ 3 dB for $n_{sp} = 1$

EDFA Noise


Output power [10dB/div]
1500
1510
1520
1530
1540
1550
1560
1570
1580
Wavelength [nm]



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Concept Test

Concept test

How is the SNR changed when doubling the input power into the optical receiver?



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A SNR reduces by 3 dB

B SNR increases by 2 dB

C SNR increases by 3 dB

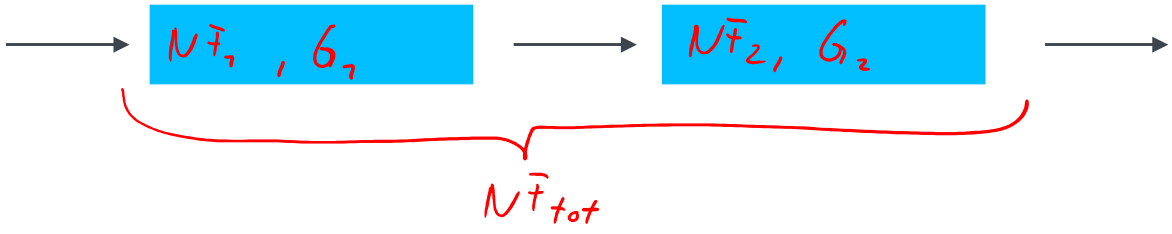
https://ilias3.uni-stuttgart.de/vote/NKE2

Noise in long distance transmission

2. Friis formula for two blocks

for Stuttgart

For two cascaded stages:



$$N\bar{F}_{Tot} = N\bar{F}_1 + \frac{N\bar{F}_2 - 1}{G_1}$$

Noise in long distance transmission

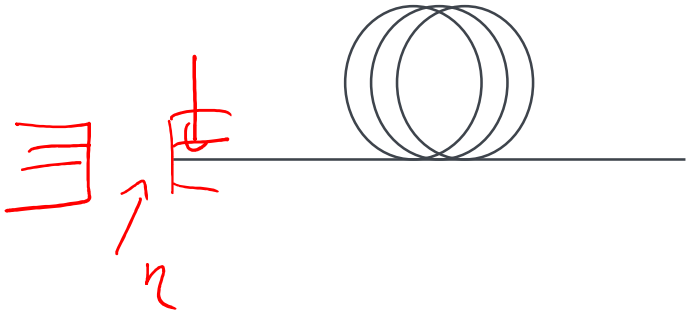
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Faserverstärker mit der Rauschzahl NF_1

Koppelwirkungsgrad η und damit einer Rauschzahl $1/\eta$ be

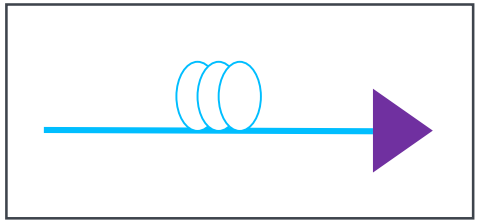
$$NF_2 = 2n_{sp}\left(1-\frac{1}{G}\right)+\frac{1}{G}$$

$$\overline{NF_1} = \frac{1}{G_1}$$

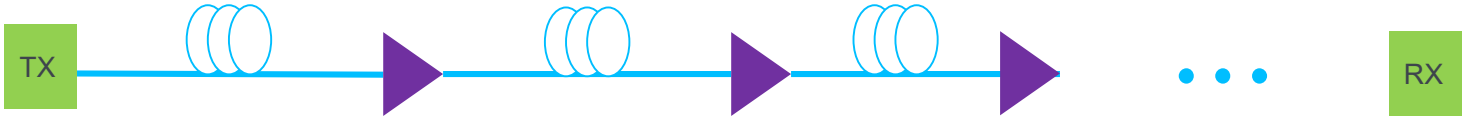


$$NF_{total} = \frac{1}{\eta} + \frac{NF_1 - 1}{\eta} = \frac{NF_1}{\eta}$$

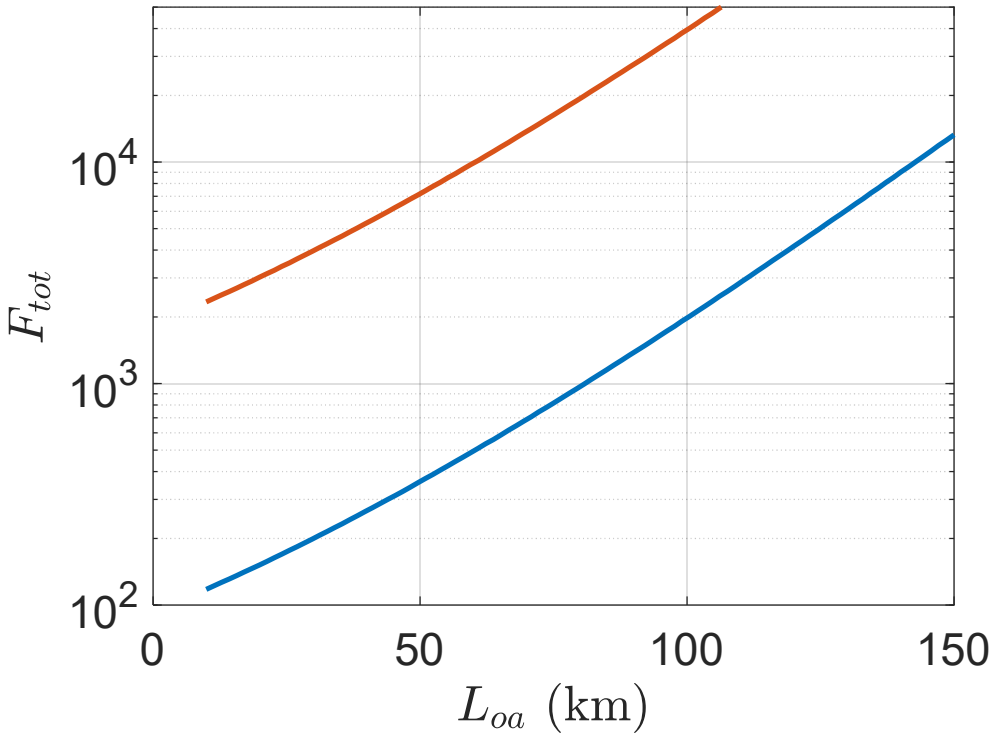
Noise in long distance transmission



Noise in long distance transmission



Noise in long distance transmission



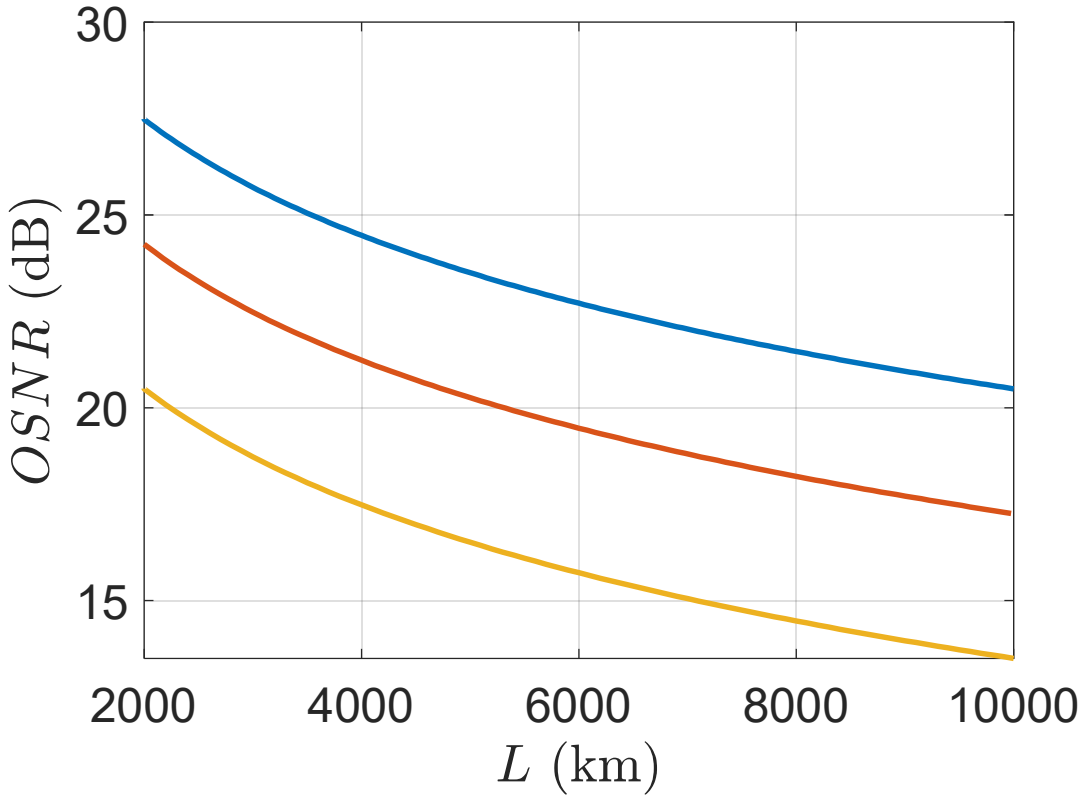
Optical Signal to Noise Ratio (OSNR)



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OSNR in long distance transmission

OSNR in long distance transmission



University of Stuttgart
Institute of electrical and optical communications

Thank you!



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phone +49 (0) 711 6856 7901
www. int.uni-stuttgart.de/

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